Name: Grisha P
Roll No: 241901030

Experiment No. 17

Create a Rogue Access Point to Capture Data Packets

AIM

To demonstrate the risk of rogue access points by setting up an unauthorized Wi-Fi access point in a controlled lab environment and capturing data packets from a client device that connects to it.

PROCEDURE

1. Set up a wireless adapter capable of AP mode on the attacker machine for creating the rogue access point.
2. Use a tool such as airbase-ng (Aircrack-ng suite) or hostapd to configure and launch a rogue access point with an SSID similar to a legitimate/trusted network (airbase-ng -e "Free_WiFi" -c 6 wlan0mon).
3. Configure IP forwarding and a DHCP server (e.g., dnsmasq) on the attacker machine so that connecting clients receive an IP address and internet/route access through the rogue AP.
4. Start Wireshark (or tcpdump) on the attacker machine's bridge/AP interface to passively capture all traffic passing through the rogue access point.
5. In the controlled lab, have a test victim device scan for and connect to the rogue access point 'Free_WiFi'.
6. Have the victim device perform typical activity (e.g., browsing an unencrypted HTTP test page or sending a test login form) while the capture is running.
7. Analyze the captured packets in Wireshark to identify the victim's traffic, and document any plaintext credentials or sensitive data observed, along with the resulting security implications.

SAMPLE OUTPUT

```
$ airbase-ng -e "Free_WiFi" -c 6 wlan0mon
Created tap interface at0
Broadcasting beacon frames for 'Free_WiFi' on channel 6...

dnsmasq log: DHCPACK 10.0.0.5 to victim-laptop (MAC A4:5E:60:AB:12:34)

Wireshark capture on at0 - victim connected and browsed test HTTP site:
Frame 88  HTTP  POST /login HTTP/1.1  testsite.local
          Form data: username=victim_user&password=Passw0rd123

Finding: Plaintext credentials captured because the test site used
unencrypted HTTP; traffic over HTTPS test sites was unreadable (expected).
```

RESULT

A rogue access point was successfully created and a test client was made to connect to it, after which data packets were captured in Wireshark, demonstrating that unencrypted traffic (HTTP) can be intercepted and credentials exposed, while encrypted HTTPS traffic remained protected.